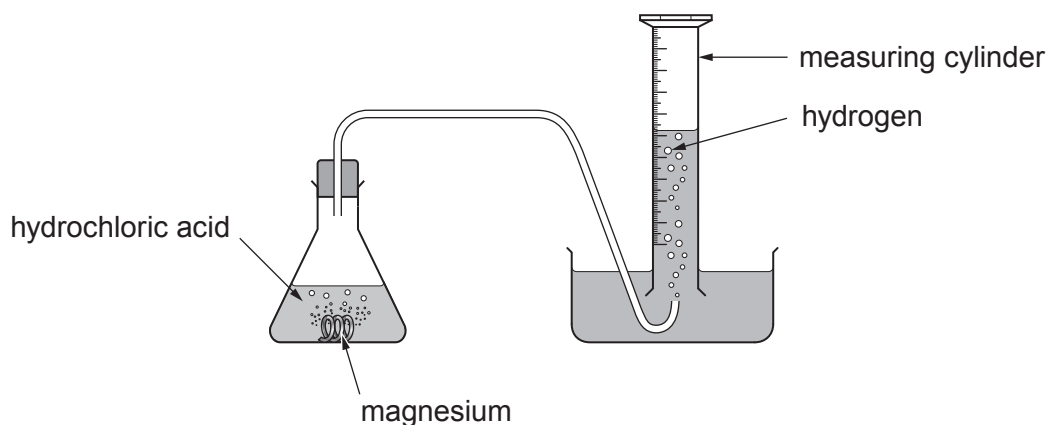
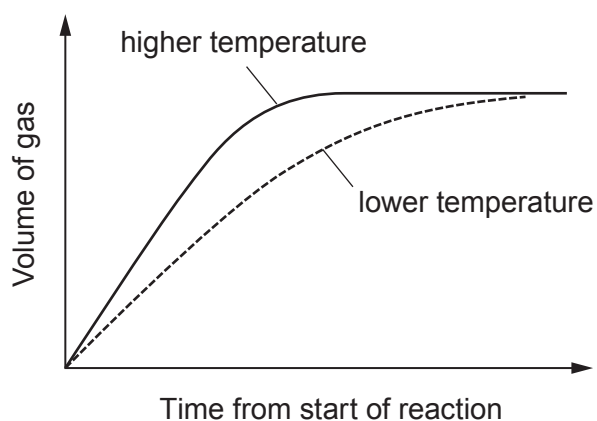


8. The following apparatus can be used to investigate the rate of the reaction between magnesium and excess dilute hydrochloric acid.



The results obtained at two different temperatures are shown below.



- (a) Explain the results obtained at different temperatures in terms of particle theory. [3]

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(b) Explain the change in rate over time. [2]

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(c) State **one** way of improving the validity of the results obtained. Explain your answer. [2]

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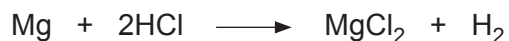
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(d) The equation for the reaction taking place is as follows.



(i) In the experiment, 0.445 g of magnesium was used.

Calculate the mass of hydrogen that will be produced during the reaction. [3]

$$A_r(\text{Mg}) = 24$$

$$A_r(\text{H}) = 1$$

Mass = g

(ii) It is known that one mole of gas has a volume of 24 dm³.

Use the equation below to calculate the volume of hydrogen produced. [2]

$$\text{number of moles} = \frac{\text{volume of gas}}{24}$$

Volume = dm³

